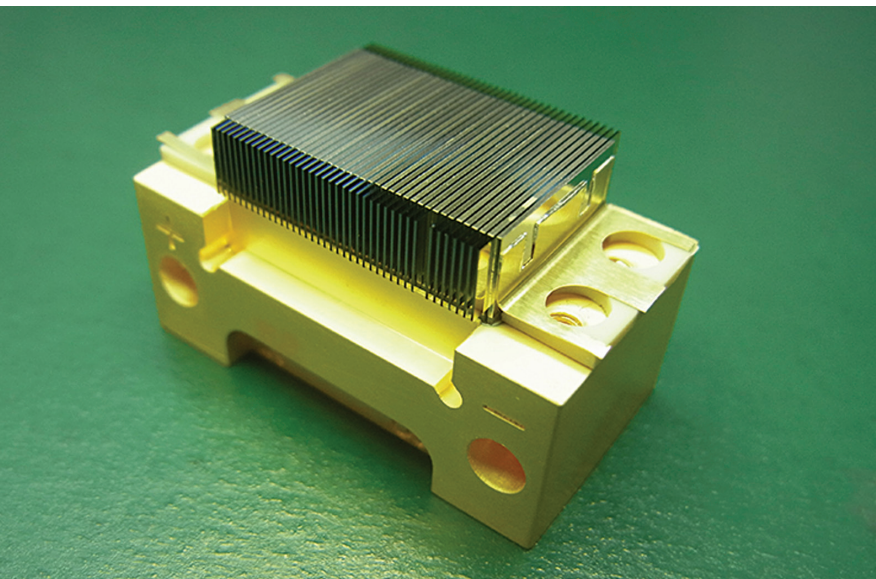


New technology for better laser modules



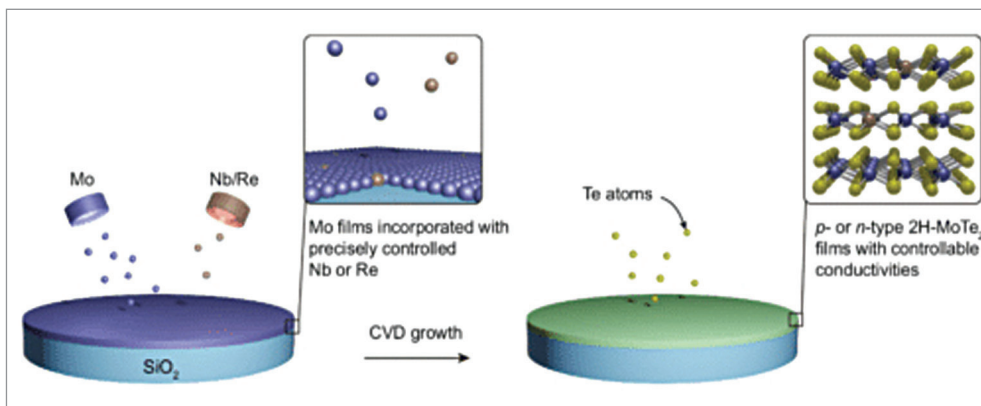
Hamamatsu Photonics has developed proprietary technology to enhance the manufacturing precision of semiconductor laser (LD) modules, enabling a new LD module with a power density of 23 kW/cm^2 —four times higher than previous models. The company claims this is the world's highest power density for laser fusion pump sources. Using this technology, modules support laser power outputs in the one-kilojoule range, marking a significant milestone in advancing laser fusion. The technology also facilitates compact LD module designs for solid-state laser pumping.

Laser diode module (Credit: Hamamatsu Photonics)

2D semiconductors transform ICs

Researchers at Peking University have developed a precise method for doping 2D semiconductors, advancing integrated circuit technology. Using substitutional doping, they synthesised 2H-MoTe₂ thin films with P-type and N-type regions, enabling chip-sized 2D CMOS inverter arrays. Unlike bulk doping methods like ion implantation, their approach leverages magnetron sputtering and chemical vapour deposition to incorporate niobium (Nb) and rhenium (Re) atoms into atomically thin films. Hall measurements confirmed Nb-doped films exhibit P-type behaviour, while Re-doped films display ambipolar or N-type

properties. Scalable to large-scale production, the method creates stable, uniform devices suitable for ultra-compact, energy-efficient electronics, including artificial intelligence and quantum computing.



Schematic diagram of incorporating the Mo film with dopant Nb or Re first by co-sputtering and then tellurising to the doped 2H-MoTe₂. The growth method is chemical vapour deposition (Credits: Peking University/Springer Nature)